Data Science Assignment

Assignment 1:

```
import pandas as pd

dataset1 = pd.DataFrame(
    {
        "StudentID": [201, 202, 203, 204, 205],
        "Name": ["Arjun", "Priya", "Rahul", "Ananya", "Vikram"],
        "Department": ["CSE", "ECE", "ME", "IT", "CIVIL"],
    }
)

dataset2 = pd.DataFrame(
    {
        "ID": [201, 202, 204, 206, 207],
        "Course": ["ML", "DBMS", "OS", "AI", "Networks"],
        "Grade": ["A", "B+", "A-", "C", "B"],
    }
)

print("Dataset 1:")
print(dataset1)
print("\nDataset 2:")
print(dataset2)

dataset2 = dataset2.rename(columns={"ID": "StudentID"})

merged_inner = pd.merge(dataset1, dataset2, on="StudentID", how="inner")
print("\nInner Merge (only matching records):")
print(merged_inner)

merged_outer = pd.merge(dataset1, dataset2, on="StudentID", how="outer")
print("\nOuter Merge (all records with NaN for missing):")
print(merged_outer)

merged_outer = merged_outer.fillna(
    {
        "Name": "Unknown",
        "Department": "Unregistered",
```

```
        "Course": "Not Enrolled",
        "Grade": "N/A",
    }
)

merged_outer["StudentID"] = merged_outer["StudentID"].astype(int)
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    POSTMAN CONSOLE

PS S:\Data Science> python -u "s:\Data Science\tempCodeRunnerFile.py"
Dataset 1:
   StudentID    Name Department
0        201   Arjun        CSE
1        202   Priya        ECE
2        203   Rahul         ME
3        204  Ananya         IT
4        205  Vikram      CIVIL

Dataset 2:
    ID    Course Grade
0  201        ML     A
1  202      DBMS    B+
2  204        OS    A-
3  206        AI     C
4  207  Networks     B

Inner Merge (only matching records):
   StudentID    Name Department Course Grade
0        201   Arjun        CSE     ML     A
1        202   Priya        ECE   DBMS    B+
2        204  Ananya         IT     OS    A-

Outer Merge (all records with NaN for missing):
   StudentID    Name Department    Course Grade
0        201   Arjun        CSE        ML     A
1        202   Priya        ECE      DBMS    B+
2        203   Rahul         ME       NaN   NaN
3        204  Ananya         IT        OS    A-
4        205  Vikram      CIVIL       NaN   NaN
5        206     NaN        NaN        AI     C
6        207     NaN        NaN  Networks     B
PS S:\Data Science>
```

Assignment 2:

```
import pandas as pd
import numpy as np

data = [
    {
        "name": "arjun",
        "email": "arjun@example.com",
        "age": 25,
        "purchaseamount": 850.0,
        "date": "2024-01-10",
        "source": "Google",
    },
    {
        "name": "priya",
        "email": "",
        "age": 30,
        "purchaseamount": 1200.0,
        "date": "2024-02-15",
        "source": "Facebook",
    },
    {
        "name": "rahul",
        "email": "rahul@example.com",
        "age": "",
        "purchaseamount": 450.5,
        "date": "2024-03-05",
        "source": "Direct",
    },
]

df = pd.DataFrame(data)

df.columns = df.columns.str.strip().str.lower()

df.drop_duplicates(inplace=True)

df["email"] = df["email"].replace("", np.nan)

df["age"] = pd.to_numeric(df["age"], errors="coerce")
```

```python
df["purchaseamount"] = pd.to_numeric(df["purchaseamount"],
errors="coerce")

df["date"] = pd.to_datetime(df["date"], errors="coerce")

df["email"] = df["email"].fillna("unknown@email.com")
df["age"] = df["age"].fillna(df["age"].median())
df["purchaseamount"] =
df["purchaseamount"].fillna(df["purchaseamount"].mean())

df = df[df["purchaseamount"] < df["purchaseamount"].quantile(0.99)]

df["name"] = df["name"].str.title()
df["source"] = df["source"].str.lower()

df["spending_category"] = np.where(df["purchaseamount"] > 1000, "High",
"Low")

analysis = df.groupby("source")["purchaseamount"].agg(["count", "mean",
"sum"])

print(df)
print("\nAnalysis by source:")
print(analysis)
```

Output:

```
PS S:\Data Science> python -u "s:\Data Science\Assignment2.py"
    name              email   age  purchaseamount       date  source spending_category
0  Arjun  arjun@example.com  25.0           850.0 2024-01-10  google               Low
2  Rahul  rahul@example.com  27.5           450.5 2024-03-05  direct               Low

Analysis by source:
        count   mean    sum
source
direct      1  450.5  450.5
google      1  850.0  850.0
PS S:\Data Science>
```

Assignment 3:

CSV FILE:

```
Lab_ID,Patient_ID,Test_Name,Result
301,1,Blood,5.5
302,2,Sugar,
303,3,Blood,6.2
304,4,Sugar,7.1
```

```
Patient_ID,Name,Age,Gender,City
1,Alice,25,Female,chennai
2,Bob,,male,delhi
3,Charlie,35,FEMALE,mumbai
4,David,40,Male,kolkata
```

```
Visit_ID,Patient_ID,Visit_Date,Doctor
201,1,2024-01-10,Dr.Smith
202,2,2024-02-15,Dr.John
203,1,2024-03-01,Dr.Smith
204,3,2024-03-20,Dr.Roy
```

Code:

```
import pandas as pd
import numpy as np

patients = pd.read_csv("patients.csv")
visits = pd.read_csv("visits.csv")
labs = pd.read_csv("labs.csv")

print(patients.info())
print(visits.info())
print(labs.info())

print(patients.isnull().sum())
print(visits.isnull().sum())
print(labs.isnull().sum())

patients["age"] = pd.to_numeric(patients["Age"], errors="coerce")
patients["age"].fillna(patients["age"].mean(), inplace=True)

labs["Result"] = pd.to_numeric(labs["Result"], errors="coerce")
labs["Result"].fillna(labs["Result"].mean(), inplace=True)
```

```
patients["Gender"] = patients["Gender"].str.capitalize()
patients["City"] = patients["City"].str.capitalize()

merged = pd.merge(patients, visits, on="Patient_ID", how="outer")
merged = pd.merge(merged, labs, on="Patient_ID", how="outer")

pivot_table = merged.pivot_table(values="Result", index="Patient_ID",
columns="Test_Name", aggfunc="mean")

melted = pivot_table.reset_index().melt(id_vars="Patient_ID",
var_name="Test_Name", value_name="Result")

visit_count =
merged.groupby("Patient_ID")["Visit_ID"].count().reset_index(name="Total_V
isits")

print(merged)
print(pivot_table)
print(melted)
print(visit_count)

merged.to_csv("final_hospital_data.csv", index=False)
```

Output:

```
PS S:\Data Science> python -u "s:\Data Science\Assignment3.py"
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 5 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Patient_ID  4 non-null      int64
 1   Name        4 non-null      object
 2   Age         3 non-null      float64
 3   Gender      4 non-null      object
 4   City        4 non-null      object
dtypes: float64(1), int64(1), object(3)
memory usage: 288.0+ bytes
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Visit_ID    4 non-null      int64
 1   Patient_ID  4 non-null      int64
 2   Visit_Date  4 non-null      object
 3   Doctor      4 non-null      object
dtypes: int64(2), object(2)
memory usage: 256.0+ bytes
None
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Lab_ID      4 non-null      int64
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    POSTMAN CONSOLE

   Patient_ID     Name  ...  Test_Name    Result
0           1    Alice  ...      Blood  5.500000
1           1    Alice  ...      Blood  5.500000
2           2      Bob  ...      Sugar  6.266667
3           3  Charlie  ...      Blood  6.200000
4           4    David  ...      Sugar  7.100000

[5 rows x 12 columns]
Test_Name   Blood     Sugar
Patient_ID
1             5.5       NaN
2             NaN  6.266667
3             6.2       NaN
4             NaN  7.100000
   Patient_ID Test_Name    Result
0           1     Blood  5.500000
1           2     Blood       NaN
2           3     Blood  6.200000
3           4     Blood       NaN
4           1     Sugar       NaN
5           2     Sugar  6.266667
6           3     Sugar       NaN
7           4     Sugar  7.100000
   Patient_ID  Total_Visits
0           1             2
1           2             1
2           3             1
3           4             0
PS S:\Data Science>
```